Nirbhay Singh

Site Reliability Engineer | Cloud Infrastructure & Observability Architect

11+	11,000+	1,000+	99.99%	14+
Years Experience	Servers Managed	Containers	Uptime Target	Certifications

PROFESSIONAL SUMMARY

Site Reliability Engineer and Cloud Architect with 11+ years ensuring production reliability across AWS, GCP, and Azure. Expert in observability (Prometheus, Grafana, ELK), Kubernetes operations (EKS, GKE, AKS), SLO/SLI-driven engineering, incident response, and disaster recovery. Managed infrastructure serving 11,000+ endpoints at PTC, orchestrated 1,000+ containers at Deloitte, and currently architects multi-cloud platforms at Bosch Europe. Builds open-source tools for proactive reliability: security scanning (ShieldIaC) and cost anomaly detection (InfraCents, TokenMeter).

KEY ACHIEVEMENTS

- 11+ years ensuring production reliability across AWS, GCP, and Azure in enterprise environments (Bosch, McAfee, Deloitte, PTC, IBM)
- Built comprehensive observability stacks: Prometheus, Grafana, ELK, CloudWatch -- unified metrics, logging, alerting, and distributed tracing
- Managed production Kubernetes clusters (EKS, GKE, AKS) -- orchestrated 1,000+ containers with automated scaling, health monitoring, and self-healing
- Designed disaster recovery strategies with cross-region replication, automated failover, and validated RTO/RPO testing
- Managed 11,000+ server infrastructure at PTC with Active Directory, high availability, and automated failover -- 99.99% uptime SLA
- P1/P2 incident response leadership across entire career: escalation management, root cause analysis, blameless post-mortems
- Built ShieldIaC (open-source): proactive IaC security scanning preventing production incidents at the infrastructure-as-code layer

PROFESSIONAL EXPERIENCE

Cloud & AI Architect (Platform Reliability)

Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Architected reliable multi-cloud platforms on GCP and AWS for Bosch Europe with 99.99% uptime targets and automated incident detection
- Designed GKE and Cloud Run platforms with pod disruption budgets, autoscaling, health probes, liveness checks, and automated rollback
- Established observability standards: structured logging, distributed tracing (Cloud Trace), SLO-based alerting, and error budget tracking
- Implemented proactive reliability: FinOps anomaly detection, security scanning (ShieldIaC), and infrastructure drift monitoring
- Mentored 20+ engineers on SRE best practices, incident response processes, blameless post-mortems, and reliability culture
- Defined SLO/SLI framework for platform services -- error budget-driven release management and change velocity governance

Senior DevOps Consultant (Platform Reliability)

Jun 2021 - Nov 2022

McAfee

- Built production observability stack: Prometheus metrics, Grafana dashboards, ELK logging, custom alerting rules, and on-call runbooks
- Deployed and operated production Kubernetes clusters across GKE, EKS, AKS, and ECS with automated node patching and security updates
- Automated infrastructure with Terraform, reducing manual toil by 50%+ and eliminating configuration drift as a reliability risk
- Developed incident response runbooks, automated remediation scripts, and escalation procedures for common failure modes

Nirbhay Singh

SRE | Observability | Kubernetes
Incident Response | Disaster Recovery

CONTACT

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CORE SKILLS

OBSERVABILITY

Prometheus, Grafana, ELK Stack, CloudWatch, Cloud Monitoring, X-Ray, Datadog

SRE PRACTICES

SLO/SLI/Error Budgets, Incident Response, Post-Mortems, Capacity Planning, Chaos Engineering, Toil Reduction

KUBERNETES

EKS, GKE, AKS, Docker, Helm, Anthos, Istio Service Mesh

CLOUD PLATFORMS

AWS, GCP, Azure, Multi-Cloud, Landing Zones

IAC & AUTOMATION

Terraform, CloudFormation, Ansible, Puppet, Python, Bash, PowerShell

CI/CD & DEPLOYS

GitHub Actions, Jenkins, ArgoCD, Canary, Blue-Green, Rolling

DR & HA

Cross-Region Replication, Automated Failover, RTO/RPO, Backup, Multi-AZ

SECURITY

IAM, Zero-Trust, ShieldIaC, OPA/Gatekeeper, Compliance

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Senior DevOps Engineer

Dec 2018 - Jun 2021

Deloitte

- Orchestrated 1,000+ containers using AWS Batch, Spot Instances, and EKS with automated health monitoring, restart policies, and graceful degradation
- Accelerated CI/CD pipeline speed by 30x while implementing reliability gates: automated testing, security scanning, and canary analysis
- Implemented multi-cloud security controls and monitoring spanning GCP, AWS, Azure, and OCI with centralised audit logging
- Reduced AWS infrastructure costs by $45,000 while maintaining SLO compliance through smart resource optimisation

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Managed 11,000+ server Windows infrastructure with Active Directory, high availability clustering, and 99.99% uptime SLA
- Designed disaster recovery: cross-region replication, automated failover, backup validation, and regular DR testing with documented RTO/RPO
- Built fault-tolerant AWS architectures (EC2, RDS Multi-AZ, S3, VPC) for production workloads with multi-AZ deployments
- Led zero-downtime cloud migrations with automated rollback strategies, canary traffic shifting, and comprehensive smoke testing
- Built operations automation using PowerShell and Ansible, reducing MTTR for common incidents by 60%+

System Engineer

Mar 2013 - Jul 2014

IBM

- P1/P2 incident response, escalation management, and root cause analysis for enterprise production systems
- Infrastructure operations, change management, and migration planning in high-availability environments

OPEN-SOURCE PROJECTS

ShieldIaC -- Proactive IaC Security Scanner

>> github.com/nirbhays/shieldiac

Prevents production security incidents at the IaC layer. Scans Terraform/CloudFormation for misconfigs. 100+ rules, 9 compliance frameworks (CIS, SOC2, HIPAA, PCI-DSS, NIST). GPT-4.1 fix suggestions.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

InfraCents -- Infrastructure Cost Anomaly Prevention

>> github.com/nirbhays/infracents

Prevents cost-related reliability issues by showing exact infrastructure cost impact of Terraform changes before merge. Catches unintended resource scaling.

STACK: Python, Next.js, Terraform, GitHub App, AWS/GCP Pricing APIs

TokenMeter -- LLM Performance & Cost Monitoring

>> github.com/nirbhays/tokenmeter

Monitors LLM latency, token usage, and costs across providers. Smart routing ensures performance SLOs while reducing spend by 60%. <5ms overhead.

STACK: Python, OpenAI, Anthropic, Google AI, Smart Routing, PyPI

CERTIFICATIONS & CREDENTIALS

- **Google** Professional Cloud DevOps Engineer (Valid Dec 2027)
- **Google** Professional Cloud Architect (Valid Jun 2026)
- **AWS** Solutions Architect - Associate (Jan 2021)
- **Google** Professional ML Engineer (Valid Dec 2027)
- **MCSA** - Microsoft Certified Solutions Associate (Apr 2015)
- **CKAD** - Certified Kubernetes Application Developer (Valid Sep 2026)
- **AWS** Solutions Architect - Professional (Feb 2020)
- **HashiCorp** Terraform Associate
- **Google** Generative AI Leader (Valid Aug 2028)